

Lab: JavaScript Events and Animations

Course: Web Design

Instructor: Mohammad Zafar Shafaq

Introduction

In this lab, students will explore one of the most important aspects of JavaScript: Event Handling and Animation. Event handling allows web pages to respond to user actions such as clicks, keyboard input, mouse movement, and page loading. Animation enables developers to create dynamic and interactive effects that improve user experience.

Task 1: Theme Switcher (Click Event)

Scenario

A university website wants users to switch between light and dark modes.

Requirements

Create:

- A heading
- A button

When the button is clicked:

- Change background color.
- Change text color.

Events Used (onclick)

Task 2: Product Image Zoom (Mouse Events)

Scenario

An online shopping website enlarges product images when users hover over them.

Requirements

Create an image.

When the mouse moves over the image:

- Increase image width.

When the mouse leaves:

- Return to original size.

Events Used: (onmouseover, onmouseout)

Task 3: Registration Form Validation

Scenario

An online registration system provides visual feedback while users fill out forms.

Requirements

Create two textboxes:

- **Username**
- **Email**

When textbox gets focus:

- Change background color to yellow.

When focus is lost:

- Restore white color.

Events Used: onfocus, onblur

Task 4: Live Password Strength Indicator

Scenario

A signup page wants to inform users about password length while typing.

Requirements

Create:

- Password textbox
- Message area

While typing:

Display:

- Weak Password (less than 8 characters)
- Strong Password (8 characters or more)

Events Used: onkeyup

Task 5: Country Selection System

Scenario: A visa application website displays selected country information.

Requirements

Create a dropdown menu containing:

- Afghanistan
- India
- Turkey
- Germany

When the user changes the selected country:

Display:

You selected Germany

(or whichever country is selected)

Events Used: onchange

Challenge Task:

Online Food Delivery Advertisement Banner

Scenario

A restaurant website wants to attract visitors with animated advertisements.

Requirements

Create:

Banner

50% Discount on Pizza

Buttons

- Start Animation
- Stop Animation

Animation Behavior

When Start Animation is clicked:

The banner should:

- Move from left to right.
- Change colors continuously.
- Increase and decrease font size.
- Repeat infinitely.

When Stop Animation is clicked:

The movement stops.

Concepts Required

`setInterval()`
`clearInterval()`
`style.left`
`style.color`
`style.fontSize`

Submission:

Students should show the progress of work on the day of LAB with IS department LAB Technician, before 3:00 PM. The complete project will be checked in coming class session on next week.